

LQR

一般形式的离散有限时间LQR问题

假设状态变量为 $x_k \in \mathbb{R}^n, k = 0, 1, \dots, N$, 控制变量为 $u_k \in \mathbb{R}^p, k = 0, 1, \dots, N-1$,

目标函数 (含有 x, u 的线性项和 x, u 的交叉项)

$$J = \sum_{k=0}^{N-1} l_k(x_k, u_k) + l_N(x_N) \quad (1)$$

其中,

$$l_k(x, u) = \frac{1}{2} x^T Q_k x + x^T S_k u + \frac{1}{2} u^T R_k u + x^T q_k + u^T r_k \quad (2)$$

$$l_N(x) = \frac{1}{2} x^T Q_N x + x^T q_N \quad (3)$$

$$Q_k \succeq 0 \quad \text{positive semi-definite} \quad (k = 1, \dots, N)$$

$$R_k \succ 0 \quad \text{positive definite} \quad (k = 1, \dots, N-1)$$

离散的状态空间方程 (含有常数项)

$$x_{k+1} = A_k x_k + B_k u_k + b_k, \quad k = 0, 1, \dots, N-1 \quad (4)$$

那么一般形式的离散有限时间LQR问题可以表达为

$$J = \sum_{k=0}^{N-1} l_k(x_k, u_k) + l_N(x_N) \quad (5)$$
$$s.t. \quad x_{k+1} = A_k x_k + B_k u_k + b_k, \quad k = 0, 1, \dots, N-1$$

目标函数(1)可以展开为

$$J = \sum_{k=0}^{N-1} \left(\frac{1}{2} x_k^T Q_k x_k + x_k^T S_k u_k + \frac{1}{2} u_k^T R_k u_k + x_k^T q_k + u_k^T r_k \right) + \frac{1}{2} x_N^T Q_N x_N + x_N^T q_N \quad (6)$$

一般形式的离散有限时间LQR问题可以展开写为

$$J = \sum_{k=0}^{N-1} \left(\frac{1}{2} x_k^T Q_k x_k + x_k^T S_k u_k + \frac{1}{2} u_k^T R_k u_k + x_k^T q_k + u_k^T r_k \right) + \frac{1}{2} x_N^T Q_N x_N + x_N^T q_N \quad (7)$$
$$s.t. \quad x_{k+1} = A_k x_k + B_k u_k + b_k, \quad k = 0, 1, \dots, N-1$$

动态规划法求解

具体推导

首先定义值函数

$$V_k(x) = \frac{1}{2} x^T P_k x + x^T p_k + c_k \quad (8)$$

下面利用动态规划法从后向前求解

对于最后一步, 目标函数(1)就是

$$J_{N \rightarrow N} = l_N(x_N) = \frac{1}{2} x_N^T Q_N x_N + x_N^T q_N \quad (9)$$

这时, 目标函数中不含有 u , 因此 $l_N(x_N)$ 就是这时的最小代价。

所以值函数 $V_N(x_N)$ 为

$$V_N(x_N) = l_N(x_N)$$

对比(3)和(8), 可知值函数 $V_N(x_N)$ 的系数分别为

$$P_N = Q_N, \quad p_N = q_N, \quad c_N = 0 \quad (10)$$

对于第 k 步到第 $k+1$ 步, 假设第 $k+1$ 步的值函数 $V_{k+1}(x)$ 已经求出, 那么第 k 步到第 $k+1$ 步的目标函数为

$$\begin{aligned}
J_{k \rightarrow k+1} &= l_k(x_k) + V_{k+1}(x_{k+1}) \\
&= l_k(x_k) + V_{k+1}(A_k x_k + B_k u_k + b_k) \\
&= \left(\frac{1}{2} x_k^T Q_k x_k + x_k^T S_k u_k + \frac{1}{2} u_k^T R_k u_k + x_k^T q_k + u_k^T r_k \right) \\
&\quad + \left[\frac{1}{2} (A_k x_k + B_k u_k + b_k)^T P_{k+1} (A_k x_k + B_k u_k + b_k) + (A_k x_k + B_k u_k + b_k)^T p_{k+1} + c_{k+1} \right] \\
&= \frac{1}{2} x_k^T (Q_k + A_k^T P_{k+1} A_k) x_k + x_k^T (S_k + A_k^T P_{k+1} B_k) u_k + \frac{1}{2} u_k^T (R_k + B_k^T P_{k+1} B_k) u_k \\
&\quad + x_k^T (q_k + A_k^T P_{k+1} b_k + A_k^T p_{k+1}) + u_k^T (r_k + B_k^T P_{k+1} b_k + B_k^T p_{k+1}) + const
\end{aligned} \tag{11}$$

记

$$\tilde{Q}_k = Q_k + A_k^T P_{k+1} A_k \tag{12}$$

$$\tilde{S}_k = S_k + A_k^T P_{k+1} B_k \tag{13}$$

$$\tilde{R}_k = R_k + B_k^T P_{k+1} B_k \tag{14}$$

$$\tilde{q}_k = q_k + A_k^T P_{k+1} b_k + A_k^T p_{k+1} \tag{15}$$

$$\tilde{r}_k = r_k + B_k^T P_{k+1} b_k + B_k^T p_{k+1} \tag{16}$$

将(12) ~ (16)代入(???)，得到

$$J_{k \rightarrow k+1} = \frac{1}{2} x_k^T \tilde{Q}_k x_k + x_k^T \tilde{S}_k u_k + \frac{1}{2} u_k^T \tilde{R}_k u_k + x_k^T \tilde{q}_k + u_k^T \tilde{r}_k + const \tag{17}$$

极小化目标函数

$$\min_u J_{k \rightarrow k+1}$$

考虑到目标函数是个连续可微的凸函数，因此对 u_k 求偏导数，并令其为0

$$\frac{\partial J_{k \rightarrow k+1}}{\partial u_k} = 0$$

得到

$$\tilde{R}_k u_k + \tilde{S}_k^T x_k + \tilde{r}_k = 0 \tag{18}$$

移项得到最优控制

$$u_k^* = -\tilde{R}_k^{-1} \tilde{S}_k^T x_k - \tilde{R}_k^{-1} \tilde{r}_k \tag{19}$$

将(13), (14), (16) 代入 (19),得到

$$\begin{aligned}
u_k^* &= - (R_k + B_k^T P_{k+1} B_k)^{-1} (S_k + A_k^T P_{k+1} B_k)^T x_k \\
&\quad - (R_k + B_k^T P_{k+1} B_k)^{-1} (r_k + B_k^T P_{k+1} b_k + B_k^T p_{k+1}) \\
&= - (R_k + B_k^T P_{k+1} B_k)^{-1} (S_k^T + B_k^T P_{k+1} A_k) x_k \\
&\quad - (R_k + B_k^T P_{k+1} B_k)^{-1} (r_k + B_k^T P_{k+1} b_k + B_k^T p_{k+1})
\end{aligned} \tag{20}$$

记

$$K_k = (R_k + B_k^T P_{k+1} B_k)^{-1} (S_k^T + B_k^T P_{k+1} A_k) \tag{21}$$

$$d_k = (R_k + B_k^T P_{k+1} B_k)^{-1} (r_k + B_k^T P_{k+1} b_k + B_k^T p_{k+1}) \tag{22}$$

那么(???)可以写为标准形式

$$u_k^* = -K_k x_k - d_k \tag{23}$$

将(19)代入回(17),得

$$\begin{aligned}
J_{k \rightarrow k+1} &= \frac{1}{2} x_k^T \tilde{Q}_k x_k + x_k^T \tilde{S}_k \left(-\tilde{R}_k^{-1} \tilde{S}_k^T x_k - \tilde{R}_k^{-1} \tilde{r}_k \right) + \frac{1}{2} \left(-\tilde{R}_k^{-1} \tilde{S}_k^T x_k - \tilde{R}_k^{-1} \tilde{r}_k \right)^T \tilde{R}_k \left(-\tilde{R}_k^{-1} \tilde{S}_k^T x_k - \tilde{R}_k^{-1} \tilde{r}_k \right) \\
&\quad + x_k^T \tilde{q}_k + \left(-\tilde{R}_k^{-1} \tilde{S}_k^T x_k - \tilde{R}_k^{-1} \tilde{r}_k \right)^T \tilde{r}_k + const \\
&= \frac{1}{2} x_k^T \left(\tilde{Q}_k - \tilde{S}_k \tilde{R}_k^{-1} \tilde{S}_k^T \right) x_k + x_k^T \left(\tilde{q}_k - \tilde{S}_k \tilde{R}_k^{-1} \tilde{r}_k \right) + const
\end{aligned} \tag{24}$$

对比值函数(8) 与 (???)，可知

$$P_k = \tilde{Q}_k - \tilde{S}_k \tilde{R}_k^{-1} \tilde{S}_k^T \tag{25}$$

$$p_k = \tilde{q}_k - \tilde{S}_k \tilde{R}_k^{-1} \tilde{r}_k \tag{26}$$

将(13), (14), (16) 代入(25)和 (26),得到对 $\forall k \in \{0, 1, \dots, N-1\}$, 有

$$P_k = Q_k + A_k^T P_{k+1} A_k - (S_k + A_k^T P_{k+1} B_k) (R_k + B_k^T P_{k+1} B_k)^{-1} (S_k + A_k^T P_{k+1} B_k)^T \quad (27)$$

$$p_k = q_k + A_k^T P_{k+1} b_k + A_k^T p_{k+1} - (S_k + A_k^T P_{k+1} B_k) (R_k + B_k^T P_{k+1} B_k)^{-1} (r_k + B_k^T P_{k+1} b_k + B_k^T p_{k+1}) \quad (28)$$

将(21), (22)代入(27), (28),可得更加简洁的形式

$$P_k = Q_k + A_k^T P_{k+1} A_k - (S_k + A_k^T P_{k+1} B_k) K_k \quad (29)$$

$$p_k = q_k + A_k^T P_{k+1} b_k + A_k^T p_{k+1} - (S_k + A_k^T P_{k+1} B_k) d_k \quad (30)$$

由此, 第 k 步的值函数 $V_k(x)$ 已经求出

$$V_k(x) = \frac{1}{2} x^T P_k x + x^T p_k + \text{const} \quad (31)$$

注意: 由于 c_k 其实在求解过程中不需要用到, 因此没有求出其显示表达式, 只是用const代替

然后, 进入第 $k-1$ 步到第 k 步的求解, 直至求解到第0步。

算法流程

给定 $A_k, B_k, b_k, R_k, S_k, r_k (k = 1, \dots, N-1)$ 和 $Q_k, q_k (k = 1, \dots, N)$

初始化 $P_N = Q_N, p_N = q_N$

反向从 $k = N-1$ 到0递推:

1. 计算 H_k
2. 计算 K_k, d_k , 见(21),(22)
3. 计算 P_k, p_k , 见(27),(28)

正向计算控制律:

$$u_k^* = -K_k x_k - d_k$$

转化为QP问题求解

将原来的状态变量为 $x_k \in \mathbb{R}^n, k = 0, 1, \dots, N$, 控制变量为 $u_k \in \mathbb{R}^p, k = 0, 1, \dots, N-1$ 分别写成大向量的形式

$$X = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \\ x_N \end{bmatrix} \in \mathbb{R}^{(N+1)n \times 1}, \quad U = \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{N-1} \end{bmatrix} \in \mathbb{R}^{Np \times 1} \quad (32)$$

目标函数的大矩阵形式

记

$$\tilde{Q} = \begin{bmatrix} Q_0 & & & & \\ & Q_1 & & & \\ & & \ddots & & \\ & & & Q_{N-1} & \\ & & & & Q_N \end{bmatrix} \succeq 0, \quad \tilde{Q} \in \mathbb{R}^{(N+1)n \times (N+1)n} \quad (33)$$

$$\tilde{R} = \begin{bmatrix} R_0 & & & \\ & R_1 & & \\ & & \ddots & \\ & & & R_{N-1} \end{bmatrix} \succ 0, \quad \tilde{R} \in \mathbb{R}^{Np \times Np} \quad (34)$$

$$\tilde{S} = \begin{bmatrix} S_0 & & & \\ & S_1 & & \\ & & \ddots & \\ & & & S_{N-1} \\ 0 & 0 & \dots & 0 \end{bmatrix}, \quad \tilde{S} \in \mathbb{R}^{(N+1)n \times Np} \quad (35)$$

$$\tilde{q} = \begin{bmatrix} q_0 \\ q_1 \\ \vdots \\ q_{N-1} \\ q_N \end{bmatrix}, \quad \tilde{q} \in \mathbb{R}^{(N+1)n \times 1} \quad (36)$$

$$\tilde{r} = \begin{bmatrix} r_0 \\ r_1 \\ \vdots \\ r_{N-1} \end{bmatrix}, \quad \tilde{r} \in \mathbb{R}^{Np \times 1} \quad (37)$$

利用(32),(33)~(37), 可以将目标函数(6)改写为

$$J = \frac{1}{2} X^T \tilde{Q} X + X^T \tilde{S} U + \frac{1}{2} U^T \tilde{R} U + X^T \tilde{q} + U^T \tilde{r} \quad (38)$$

状态空间方程的大矩阵形式

根据状态空间方程(4):

$$x_{k+1} = A_k x_k + B_k u_k + b_k$$

可以从 x_0 递推到 $x_1, x_2, \dots, x_N$

$$x_0 = I x_0 \quad (39)$$

$$x_1 = A_0 x_0 + B_0 u_0 + b_0 \quad (40)$$

$$\begin{aligned} x_2 &= A_1 x_1 + B_1 u_1 + b_1 \\ &= A_1 A_0 x_0 + A_1 B_0 u_0 + B_1 u_1 + A_1 b_0 + b_1 \end{aligned} \quad (41)$$

$$\begin{aligned} x_3 &= A_2 x_2 + B_2 u_2 + b_2 \\ &= A_2 A_1 A_0 x_0 + A_2 A_1 B_0 u_0 + A_2 B_1 u_1 + B_2 u_2 + A_2 A_1 b_0 + A_2 b_1 + b_2 \end{aligned} \quad (42)$$

$$\begin{aligned} &\vdots \\ x_N &= \left(\prod_{i=0}^{N-1} A_i \right) x_0 + \sum_{i=0}^{N-1} \left(\prod_{j=i+1}^{N-1} A_j \right) B_i u_i + \sum_{i=0}^{N-1} \left(\prod_{j=i+1}^{N-1} A_j \right) b_i \end{aligned} \quad (43)$$

Note

注意: 第一条式子 $x_0 = I x_0$ 是为了凑大矩阵, 便于后面交叉项的计算而加入的。下面就会看到它的妙用。

记

$$\tilde{A} = \begin{bmatrix} I \\ A_0 \\ A_1 A_0 \\ A_2 A_1 A_0 \\ \vdots \\ \prod_{i=0}^{N-1} A_i \end{bmatrix}, \quad \tilde{A} \in \mathbb{R}^{(N+1)n \times n} \quad (44)$$

$$\tilde{B} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ B_0 & 0 & 0 & \cdots & 0 \\ A_1 B_0 & B_1 & 0 & \cdots & 0 \\ A_2 A_1 B_0 & A_2 B_1 & B_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ (A_{N-1} \cdots A_1) B_0 & (A_{N-1} \cdots A_2) B_1 & (A_{N-1} \cdots A_3) B_2 & \cdots & B_{N-1} \end{bmatrix}, \quad \tilde{B} \in \mathbb{R}^{(N+1)n \times Np} \quad (45)$$

$$\tilde{b} = \begin{bmatrix} 0 \\ b_0 \\ A_1 b_0 + b_1 \\ A_2 A_1 b_0 + A_2 b_1 + b_2 \\ \vdots \\ (A_{N-1} \cdots A_1) b_0 + (A_{N-1} \cdots A_2) b_1 + \cdots + b_{N-1} \end{bmatrix}, \quad \tilde{b} \in \mathbb{R}^{(N+1)n \times 1} \quad (46)$$

利用(32), (44), (45), (46), 可以将从 x_0 递推到 $x_1, x_2, \dots, x_N$ 的公式(39)~(43)写成大矩阵形式

$$X = \tilde{A} x_0 + \tilde{B} U + \tilde{b} \quad (47)$$

最优控制量的求解

在实际问题中, x_0 是已知的初始状态, 而 $x_1, \dots, x_N$ 都是未知的, 故 X 是未知。所以将(47)代入目标函数(38),可以消去未知的 X , 只剩下未知数 U

$$\begin{aligned} J &= \frac{1}{2} \left(\tilde{A}x_0 + \tilde{B}U + \tilde{b} \right)^T \tilde{Q} \left(\tilde{A}x_0 + \tilde{B}U + \tilde{b} \right) + \left(\tilde{A}x_0 + \tilde{B}U + \tilde{b} \right)^T \tilde{S}U + \frac{1}{2} U^T \tilde{R}U \\ &\quad + \left(\tilde{A}x_0 + \tilde{B}U + \tilde{b} \right)^T \tilde{q} + U^T \tilde{r} \\ &= \frac{1}{2} U^T \left(\tilde{R} + \tilde{B}^T \tilde{Q} \tilde{B} + \tilde{B}^T \tilde{S} + \tilde{S}^T \tilde{B} \right) U + U^T \left[\tilde{B}^T \tilde{Q} \left(\tilde{A}x_0 + \tilde{b} \right) + \tilde{B}^T \tilde{q} + \tilde{r} + \tilde{S}^T \left(\tilde{A}x_0 + \tilde{b} \right) \right] \\ &\quad + \frac{1}{2} \left(\tilde{A}x_0 + \tilde{b} \right)^T \tilde{Q} \left(\tilde{A}x_0 + \tilde{b} \right) + \left(\tilde{A}x_0 + \tilde{b} \right)^T \tilde{q} \end{aligned} \quad (48)$$

记

$$H = \tilde{R} + \tilde{B}^T \tilde{Q} \tilde{B} + \tilde{B}^T \tilde{S} + \tilde{S}^T \tilde{B} \quad (49)$$

$$\begin{aligned} g &= \tilde{B}^T \tilde{Q} \left(\tilde{A}x_0 + \tilde{b} \right) + \tilde{B}^T \tilde{q} + \tilde{r} + \tilde{S}^T \left(\tilde{A}x_0 + \tilde{b} \right) \\ &= \left(\tilde{B}^T \tilde{Q} + \tilde{S}^T \right) \tilde{A}x_0 + \left(\tilde{B}^T \tilde{Q} + \tilde{S}^T \right) \tilde{b} + \tilde{B}^T \tilde{q} + \tilde{r} \end{aligned} \quad (50)$$

代入(???)，得到

$$J = \frac{1}{2} U^T H U + U^T g + const$$

对 U 求偏导, 并令其为0

$$\frac{\partial J}{\partial U} = 0$$

即

$$H U + g = 0$$

由于 $\tilde{R} \succ 0, \tilde{Q} \succeq 0$, 所以 $H \succ 0$, 说明 H 可逆。所以可以求出 U 为

$$U = -H^{-1}g$$

u_0^* 即为 U 的第一项

$$u_0^* = [1 \quad 0 \quad \dots \quad 0]U$$

KKT条件求解

对于一般形式的LQR问题(7)(为了方便查阅, 将它在此处再写一次)

$$\begin{aligned} J &= \sum_{k=0}^{N-1} \left(\frac{1}{2} x_k^T Q_k x_k + x_k^T S_k u_k + \frac{1}{2} u_k^T R_k u_k + x_k^T q_k + u_k^T r_k \right) + \frac{1}{2} x_N^T Q_N x_N + x_N^T q_N \\ s.t. \quad x_{k+1} &= A_k x_k + B_k u_k + b_k, \quad k = 0, 1, \dots, N-1 \end{aligned}$$

KKT条件

通过引入拉格朗日乘子 $\lambda_k \in \mathbb{R}^n$, $k = 1, 2, \dots, N$, 写出其拉格朗日函数

$$\begin{aligned} L(x, u, \lambda) &= \sum_{k=0}^{N-1} \left[\frac{1}{2} x_k^T Q_k x_k + x_k^T S_k u_k + \frac{1}{2} u_k^T R_k u_k + x_k^T q_k + u_k^T r_k + \lambda_{k+1}^T (A_k x_k + B_k u_k + b_k - x_{k+1}) \right] \\ &\quad + \frac{1}{2} x_N^T Q_N x_N + x_N^T q_N \end{aligned} \quad (51)$$

对 u_k 求偏导 $k = 0, 1, \dots, N-1$, 并令其为0

$$\frac{\partial L(x, u, \lambda)}{\partial u_k} = 0$$

即

$$S_k^T x_k + R_k u_k + r_k + B_k^T \lambda_{k+1} = 0$$

移项得

$$u_k = -R_k^{-1} S_k^T x_k - R_k^{-1} (r_k + B_k^T \lambda_{k+1}) \quad (52)$$

对 x_k 求偏导 $k = 0, 1, \dots, N-1$, 并令其为0

$$\frac{\partial L(x, u, \lambda)}{\partial x_k} = 0$$

即

$$Q_k x_k + S_k u_k + q_k + A_k^T \lambda_{k+1} - \lambda_k = 0$$

移项得

$$\lambda_k = Q_k x_k + S_k u_k + q_k + A_k^T \lambda_{k+1} \quad (53)$$

对 x_N 求偏导, 并令其为0

$$\frac{\partial L(x, u, \lambda)}{\partial x_N} = 0$$

即

$$Q_N x_N + q_N - \lambda_N = 0$$

移项得

$$\lambda_N = Q_N x_N + q_N \quad (54)$$

(53)和(54)称为协态方程, 其中(54)是协态方程的终止条件。

综上, 可以得到KKT条件

KKT条件由状态空间方程, 协态方程, 和控制最优性条件构成。

The KKT conditions lead to the state space equation, the costate equation, and the optimality condition.

$$\text{Primal feasible: } x_{k+1} = A_k x_k + B_k u_k + b_k, \quad k = 0, 1, \dots, N-1 \quad (55)$$

$$\text{Dual feasible: not exists}$$

$$\text{Complementarity: not exists}$$

$$\text{Stationarity: } \begin{cases} u_k = -R_k^{-1} S_k^T x_k - R_k^{-1} (r_k + B_k^T \lambda_{k+1}), & k = 0, 1, \dots, N-1 \\ \lambda_k = Q_k x_k + S_k u_k + q_k + A_k^T \lambda_{k+1}, & k = 0, 1, \dots, N-1 \\ \lambda_N = Q_N x_N + q_N. \end{cases} \quad (56)$$

最优控制量与Riccati方程的推导

不难发现, KKT条件中全部都是等式, 因此求解 u_k ($k = 0, 1, \dots, N-1$)就相当于解方程, 可以直接利用消元法求解。

当 $k = N-1$ 时, (52)就是

$$u_{N-1} = -R_{N-1}^{-1} S_{N-1}^T x_{N-1} - R_{N-1}^{-1} (r_{N-1} + B_{N-1}^T \lambda_N)$$

代入(54)的 λ_N 得

$$u_{N-1} = -R_{N-1}^{-1} S_{N-1}^T x_{N-1} - R_{N-1}^{-1} [r_{N-1} + B_{N-1}^T (Q_N x_N + q_N)]$$

代入 $k = N-1$ 时的状态空间方程

$$x_N = A_{N-1} x_{N-1} + B_{N-1} u_{N-1} + b_{N-1}$$

得到

$$u_{N-1} = -R_{N-1}^{-1} S_{N-1}^T x_{N-1} - R_{N-1}^{-1} \{r_{N-1} + B_{N-1}^T [Q_N (A_{N-1} x_{N-1} + B_{N-1} u_{N-1} + b_{N-1}) + q_N]\} \quad (57)$$

整理得

$$\begin{aligned} u_{N-1} = & - (R_{N-1} + B_{N-1}^T Q_N B_{N-1})^{-1} (S_{N-1}^T + B_{N-1}^T Q_N A_{N-1}) x_{N-1} \\ & - (R_{N-1} + B_{N-1}^T Q_N B_{N-1})^{-1} (r_{N-1} + B_{N-1}^T Q_N b_{N-1} + B_{N-1}^T Q_N q_N) \end{aligned} \quad (58)$$

记

$$K_{N-1} = (R_{N-1} + B_{N-1}^T Q_N B_{N-1})^{-1} (S_{N-1}^T + B_{N-1}^T Q_N A_{N-1}) \quad (59)$$

$$d_{N-1} = (R_{N-1} + B_{N-1}^T Q_N B_{N-1})^{-1} (r_{N-1} + B_{N-1}^T Q_N b_{N-1} + B_{N-1}^T Q_N q_N) \quad (60)$$

则

$$u_{N-1} = -K_{N-1} x_{N-1} - d_{N-1} \quad (61)$$

不难发现, 这里得到的结果与动态规划求出来的结果一致。

当 $k = N - 1$ 时, (53)就是

$$\lambda_{N-1} = Q_{N-1}x_{N-1} + S_{N-1}u_{N-1} + q_{N-1} + A_{N-1}^T \lambda_N \quad (62)$$

代入(54)的 λ_N ,得

$$\lambda_{N-1} = Q_{N-1}x_{N-1} + S_{N-1}u_{N-1} + q_{N-1} + A_{N-1}^T (Q_N x_N + q_N)$$

再代入(61)中的 u_{N-1} 和 $k = N - 1$ 时的状态空间方程

$$x_N = A_{N-1}x_{N-1} + B_{N-1}u_{N-1} + b_{N-1}$$

得到

$$\begin{aligned} \lambda_{N-1} = & Q_{N-1}x_{N-1} + S_{N-1}(-K_{N-1}x_{N-1} - d_{N-1}) \\ & + q_{N-1} + A_{N-1}^T [Q_N(A_{N-1}x_{N-1} + B_{N-1}u_{N-1} + b_{N-1}) + q_N] \end{aligned}$$

整理得

$$\begin{aligned} \lambda_{N-1} = & [Q_{N-1} + A_{N-1}^T Q_N A_{N-1} - (S_{N-1} + A_{N-1}^T Q_N B_{N-1})K_{N-1}]x_{N-1} \\ & + q_{N-1} + A_{N-1}^T q_N + A_{N-1}^T Q_N b_{N-1} - (S_{N-1} + A_{N-1}^T Q_N B_{N-1})d_{N-1} \end{aligned} \quad (63)$$

对比(54)和(???)可以发现, λ_k 可以写成关于 x_k 的仿射形式(即关于 x_k 的线性项加一个常数项), 将线性项系数即为 P_k , 常数项系数记为 p_k , 则

$$\lambda_k = P_k x_k + p_k \quad (64)$$

那么(54)就可以写成

$$P_N = Q_N, \quad p_N = q_N \quad (65)$$

$$\lambda_N = P_N x_N + p_N \quad (66)$$

同理, (???)可以写成

$$P_{N-1} = Q_{N-1} + A_{N-1}^T P_N A_{N-1} - (S_{N-1} + A_{N-1}^T Q_N B_{N-1})K_{N-1} \quad (67)$$

$$p_{N-1} = q_{N-1} + A_{N-1}^T Q_N b_{N-1} + A_{N-1}^T q_N - (S_{N-1} + A_{N-1}^T Q_N B_{N-1})d_{N-1} \quad (68)$$

$$\lambda_{N-1} = P_{N-1} x_{N-1} + p_{N-1} \quad (69)$$

像这样交替求解 u_k, λ_k , , 可以求出 $u_{N-2}, \lambda_{N-2}, \dots, u_1, \lambda_1, u_0$

⚠ Caution

注意: λ_k 只有 $k = 1, 2, \dots, N$ 这 N 项, 并没有 $k = 0$ 项!

下面, 我们直接对 k 进行推导。

由于前面已发现, λ_k 可以写成关于 x_k 的仿射形式。所以假设协态方程为

$$\lambda_k = P_k x_k + p_k, \quad k = 1, \dots, N$$

也可以写为

$$\lambda_{k+1} = P_{k+1} x_{k+1} + p_{k+1}, \quad k = 0, \dots, N-1 \quad (70)$$

代入(52), 可以得到

$$u_k = -R_k^{-1} S_k^T x_k - R_k^{-1} [r_k + B_k^T (P_{k+1} x_{k+1} + p_{k+1})], \quad k = 0, \dots, N-1$$

代入状态空间方程

$$x_{k+1} = A_k x_k + B_k u_k + b_k, \quad k = 0, 1, \dots, N-1 \quad (71)$$

可以得到

$$u_k = -R_k^{-1} S_k^T x_k - R_k^{-1} \{r_k + B_k^T [P_{k+1} (A_k x_k + B_k u_k + b_k) + p_{k+1}]\}, \quad k = 0, 1, \dots, N-1$$

整理得

$$\begin{aligned} u_k = & - (R_k + B_k^T P_{k+1} B_k)^{-1} (S_k^T + B_k^T P_{k+1} A_k) x_k \\ & - (R_k + B_k^T P_{k+1} B_k)^{-1} (r_k + B_k^T P_{k+1} b_k + B_k^T p_{k+1}) \end{aligned}, \quad k = 0, 1, \dots, N-1 \quad (72)$$

记

$$K_k = (R_k + B_k^T P_{k+1} B_k)^{-1} (S_k^T + B_k^T P_{k+1} A_k), \quad k = 0, 1, \dots, N-1 \quad (73)$$

$$d_k = (R_k + B_k^T P_{k+1} B_k)^{-1} (r_k + B_k^T P_{k+1} b_k + B_k^T p_{k+1}), \quad k = 0, 1, \dots, N-1 \quad (74)$$

代回(72),得

$$u_k = -K_k x_k - d_k, \quad k = 0, 1, \dots, N-1 \quad (75)$$

将(75),(70)和(71)代入(53),得到

$$\lambda_k = Q_k x_k + S_k (-K_k x_k - d_k) + q_k + A_k^T [P_{k+1} (A_k x_k + B_k u_k + b_k) + p_{k+1}], \quad k = 1, 2, \dots, N-1$$

整理得

$$\lambda_k = [Q_k + A_k^T P_{k+1} A_k - (S_k + A_k^T P_{k+1} B_k) K_k] x_k + q_k + A_k^T p_{k+1} + A_k^T P_{k+1} b_k - (S_k + A_k^T P_{k+1} B_k) d_k, \quad k = 1, 2, \dots, N-1 \quad (76)$$

记

$$P_k = Q_k + A_k^T P_{k+1} A_k - (S_k + A_k^T P_{k+1} B_k) K_k, \quad k = 1, 2, \dots, N-1 \quad (77)$$

$$p_k = q_k + A_k^T p_{k+1} + A_k^T P_{k+1} b_k - (S_k + A_k^T P_{k+1} B_k) d_k, \quad k = 1, 2, \dots, N-1 \quad (78)$$

代回(76),又可以得到(64)的形式

$$\lambda_k = P_k x_k + p_k, \quad k = 1, 2, \dots, N-1$$

像这样交替求解 u_k, λ_k , 直至求出 u_0

总结

利用KKT条件中的等式, 从后向前, 即从 $k = N$ 到0反推, 每次通过互相代入消元, 交替求解 u_k, λ_k , 直至求出 u_0

求解顺序依次为: $\lambda_N \rightarrow u_{N-1} \rightarrow \lambda_{N-1} \rightarrow \dots \rightarrow u_1 \rightarrow \lambda_1 \rightarrow u_0$

控制量

$$u_k = -K_k x_k - d_k, \quad k = 0, 1, \dots, N-1 \quad (79)$$

其中,

$$K_k = (R_k + B_k^T P_{k+1} B_k)^{-1} (S_k^T + B_k^T P_{k+1} A_k), \quad k = 0, 1, \dots, N-1 \quad (80)$$

$$d_k = (R_k + B_k^T P_{k+1} B_k)^{-1} (r_k + B_k^T P_{k+1} b_k + B_k^T p_{k+1}), \quad k = 0, 1, \dots, N-1 \quad (81)$$

协态方程

$$\lambda_k = P_k x_k + p_k, \quad k = 1, 2, \dots, N \quad (82)$$

其中,

$$P_k = Q_k + A_k^T P_{k+1} A_k - (S_k + A_k^T P_{k+1} B_k) K_k, \quad k = 1, 2, \dots, N-1 \quad (83)$$

$$p_k = q_k + A_k^T p_{k+1} + A_k^T P_{k+1} b_k - (S_k + A_k^T P_{k+1} B_k) d_k, \quad k = 1, 2, \dots, N-1 \quad (84)$$

边界条件为

$$P_N = Q_N, \quad p_N = q_N \quad (85)$$

块KKT系统

块KKT系统本质上跟前文的“KKT条件推导”是相同的, 都是利用了KKT条件。只是“块KKT系统”将所有变量和参数写成大向量或大矩阵, 从矩阵的视角来看待整个推导过程。

对于一般形式的LQR问题(7)(为了方便查阅, 将它在此处再写一次)

$$J = \sum_{k=0}^{N-1} \left(\frac{1}{2} x_k^T Q_k x_k + x_k^T S_k u_k + \frac{1}{2} u_k^T R_k u_k + x_k^T q_k + u_k^T r_k \right) + \frac{1}{2} x_N^T Q_N x_N + x_N^T q_N$$

$$s. t. \quad x_{k+1} = A_k x_k + B_k u_k + b_k, \quad k = 0, 1, \dots, N-1$$

下面分别将目标函数和约束条件写成大矩阵形式

目标函数的大矩阵形式

首先将所有变量写成一个大向量

$$z = \begin{bmatrix} X \\ U \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \\ x_N \\ u_0 \\ u_1 \\ \vdots \\ u_{N-1} \end{bmatrix}, \quad z \in \mathbb{R}^{[(N+1)n+Np] \times 1} \quad (86)$$

然后将所有权重矩阵写成大矩阵形式

$$\tilde{Q} = \begin{bmatrix} Q_0 & & & & \\ & Q_1 & & & \\ & & \ddots & & \\ & & & Q_{N-1} & \\ & & & & Q_N \end{bmatrix} \succeq 0, \quad \tilde{Q} \in \mathbb{R}^{(N+1)n \times (N+1)n} \quad (87)$$

$$\tilde{R} = \begin{bmatrix} R_0 & & & & \\ & R_1 & & & \\ & & \ddots & & \\ & & & R_{N-1} & \end{bmatrix} \succ 0, \quad \tilde{R} \in \mathbb{R}^{Np \times Np} \quad (88)$$

$$\tilde{S} = \begin{bmatrix} S_0 & & & & \\ & S_1 & & & \\ & & \ddots & & \\ & & & S_{N-1} & \\ 0 & 0 & \cdots & 0 & \end{bmatrix}, \quad \tilde{S} \in \mathbb{R}^{(N+1)n \times Np} \quad (89)$$

记

$$H = \begin{bmatrix} \tilde{Q} & \tilde{S} \\ \tilde{S}^T & \tilde{R} \end{bmatrix}, \quad H \in \mathbb{R}^{[(N+1)n+Np] \times [(N+1)n+Np]} \quad (90)$$

将线性项的系数也写成大向量形式

$$\tilde{q} = \begin{bmatrix} q_0 \\ q_1 \\ \vdots \\ q_{N-1} \\ q_N \end{bmatrix}, \quad \tilde{q} \in \mathbb{R}^{(N+1)n \times 1} \quad (91)$$

$$\tilde{r} = \begin{bmatrix} r_0 \\ r_1 \\ \vdots \\ r_{N-1} \end{bmatrix}, \quad \tilde{r} \in \mathbb{R}^{Np \times 1} \quad (92)$$

记

$$f = \begin{bmatrix} \tilde{q} \\ \tilde{r} \end{bmatrix} = \begin{bmatrix} q_0 \\ q_1 \\ \vdots \\ q_{N-1} \\ q_N \\ r_0 \\ r_1 \\ \vdots \\ r_{N-1} \end{bmatrix}, \quad f \in \mathbb{R}^{[(N+1)n+Np] \times 1} \quad (93)$$

所以一般形式的LQR问题(7)的目标函数可写为

$$J = \frac{1}{2} z^T H z + f^T z \quad (94)$$

约束条件的大矩阵形式

约束条件

$$x_{k+1} = A_k x_k + B_k u_k + b_k, \quad k = 0, 1, \dots, N-1$$

可以改写为

$$-A_k x_k + x_{k+1} - B_k u_k = b_k, \quad k = 0, 1, \dots, N-1$$

将这 N 个约束方程都写出来:

$$\begin{array}{ccccccc} -A_0 x_0 & + I x_1 & & & & -B_0 u_0 & = b_0 \\ & -A_1 x_1 & + I x_2 & & & -B_1 u_1 & = b_1 \\ & & & \ddots & & & \vdots \\ & & & & -A_{N-1} x_{N-1} & + I x_N & -B_{N-1} u_{N-1} = b_{N-1} \end{array} \quad (95)$$

记

$$G = \begin{bmatrix} -A_0 & I & 0 & \cdots & 0 & -B_0 & 0 & \cdots & 0 \\ 0 & -A_1 & I & \cdots & 0 & 0 & -B_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_{N-1} & I & 0 & 0 & \cdots & -B_{N-1} \end{bmatrix}, \quad G \in \mathbb{R}^{Nn \times [(N+1)n + Np]} \quad (96)$$

$$\beta = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{N-1} \end{bmatrix}, \quad \beta \in \mathbb{R}^{Nn \times 1} \quad (97)$$

则(95)可以写成矩阵形式

$$Gz = \beta \quad (98)$$

LQR问题的大矩阵形式

综上, 可以将原来的LQR问题写成大矩阵形式

$$\begin{array}{l} J = \frac{1}{2} z^T H z + f^T z \\ s. t. \quad Gz = \beta \end{array} \quad (99)$$

块KKT条件

引入拉格朗日乘子

$$\lambda = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_N \end{bmatrix}, \quad \lambda \in \mathbb{R}^{Nn \times 1} \quad (100)$$

可以写出拉格朗日函数

$$L(z, \lambda) = \frac{1}{2} z^T H z + f^T z + \lambda^T (Gz - \beta) \quad (101)$$

对 z 求梯度, 并令其为0

$$\nabla_z L(z, \lambda) = 0$$

即

$$Hz + f + \lambda^T G = 0 \quad (102)$$

所以KKT条件为

$$\text{Primal feasible: } Gz = \beta \quad (103)$$

$$\text{Dual feasible: not exists}$$

$$\text{Complementarity: not exists}$$

$$\text{Stationarity: } Hz + f + \lambda^T G = 0 \quad (104)$$

这个KKT条件也可以进一步写成矩阵形式

$$\begin{bmatrix} H & G^T \\ G & 0 \end{bmatrix} \begin{bmatrix} z \\ \lambda \end{bmatrix} = \begin{bmatrix} -f \\ \beta \end{bmatrix} \quad (105)$$

对这个矩阵方程进行求解，即可得到最优控制量。

进一步，将(105)展开

$$\begin{bmatrix} Q_0 & 0 & \cdots & 0 & 0 & S_0 & 0 & \cdots & 0 & -A_0 & 0 & \cdots & 0 \\ 0 & Q_1 & \cdots & 0 & 0 & 0 & S_1 & \cdots & 0 & I & -A_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & Q_{N-1} & 0 & 0 & 0 & \cdots & S_{N-1} & 0 & 0 & \cdots & -A_{N-1} \\ 0 & 0 & \cdots & 0 & Q_N & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ S_0 & 0 & \cdots & 0 & 0 & R_0 & 0 & \cdots & 0 & -B_0 & 0 & \cdots & 0 \\ 0 & S_1 & \cdots & 0 & 0 & 0 & R_1 & \cdots & 0 & 0 & -B_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_{N-1} & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & -B_{N-1} \\ -A_0 & I & \cdots & 0 & 0 & -B_0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & -A_1 & \cdots & 0 & 0 & 0 & -B_1 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_{N-1} & 0 & 0 & 0 & \cdots & -B_{N-1} & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \\ x_N \\ u_0 \\ u_1 \\ \vdots \\ u_{N-1} \\ \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_N \end{bmatrix} = \begin{bmatrix} q_0 \\ q_1 \\ \vdots \\ q_{N-1} \\ q_N \\ r_0 \\ r_1 \\ \vdots \\ r_{N-1} \\ b_0 \\ b_1 \\ \vdots \\ b_{N-1} \end{bmatrix} \quad (106)$$

一眼看过去，上面的矩阵非常凌乱，但是如果我们将变量 $[z \ \lambda]^T$ 按时间排序为：

$$x_0, u_0, \lambda_1, x_1, u_1, \lambda_2, \cdots, x_{N-1}, u_{N-1}, \lambda_N, x_N$$

(106)就可以写成带状矩阵的形式

$$\begin{bmatrix} Q_0 & S_0 & -A_0 & & & & & & & & & & \\ S_0 & R_0 & -B_0 & 0 & & & & & & & & & \\ -A_0 & -B_0 & 0 & I & 0 & & & & & & & & \\ & 0 & I & Q_1 & S_1 & -A_1 & & & & & & & \\ & & 0 & S_1 & R_1 & -B_1 & 0 & & & & & & \\ & & & -A_1 & -B_1 & 0 & I & 0 & & & & & \\ & & & & \ddots & \ddots & \ddots & \ddots & \ddots & & & & \\ & & & & & -A_{N-2} & -B_{N-2} & 0 & I & 0 & & & \\ & & & & & & 0 & I & Q_{N-1} & S_{N-1} & -A_{N-1} & & \\ & & & & & & & 0 & S_{N-1} & R_{N-1} & -B_{N-1} & 0 \\ & & & & & & & & -A_{N-1} & -B_{N-1} & 0 & I \\ & & & & & & & & & 0 & I & Q_N \end{bmatrix} \begin{bmatrix} x_0 \\ u_0 \\ \lambda_1 \\ x_1 \\ u_1 \\ \lambda_2 \\ \vdots \\ \lambda_{N-1} \\ x_{N-1} \\ u_{N-1} \\ \lambda_N \\ x_N \end{bmatrix} = \begin{bmatrix} q_0 \\ r_0 \\ b_0 \\ q_1 \\ r_1 \\ b_1 \\ \vdots \\ b_{N-2} \\ q_{N-1} \\ r_{N-1} \\ b_{N-1} \\ q_N \end{bmatrix} \quad (107)$$

从这个角度来看，求解LQR问题其实就是求解(107)这个矩阵方程。而从下往上消元的过程，其实就对应着前文的“KKT条件求解”的过程。